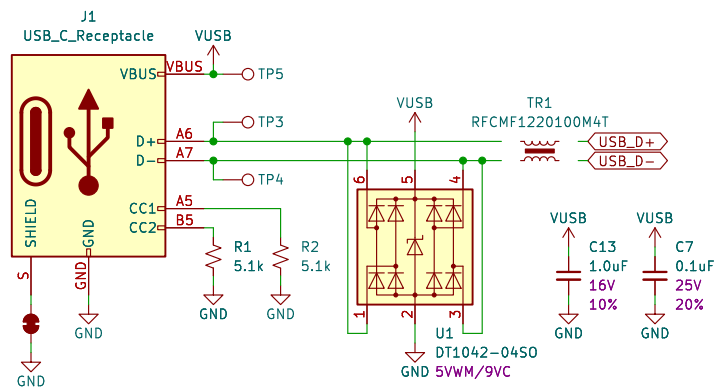
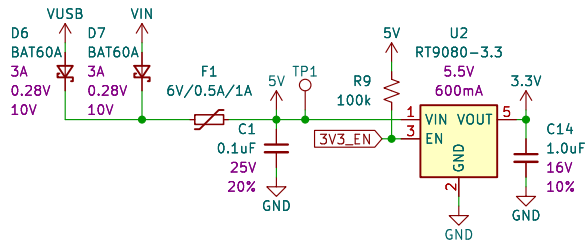


USB

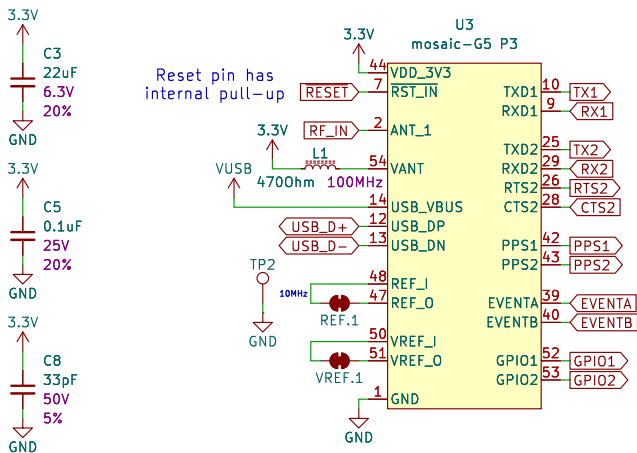


USB Track Impedance: Coplanar Differential Pair @ 90Ω
<https://jlcpcb.com/pcb-impedance-calculator>
Board Thickness: 1.6mm, Layers: 4, Er: 4.4 (7628 Prepreg)
Dielectric Thickness (Layer 1 to 2): 8.28mil/0.21mm (JLC7628)
Copper Thickness (1oz outer / 0.5oz inner): 1.38mil/0.035mm
Polygon Isolation: 7mil/0.1778mm
Trace Spacing: 9.07mil/0.2304mm
Trace Width: 10.93mil/0.2776mm
Trace center-to-center: 20.0mil/0.508mm

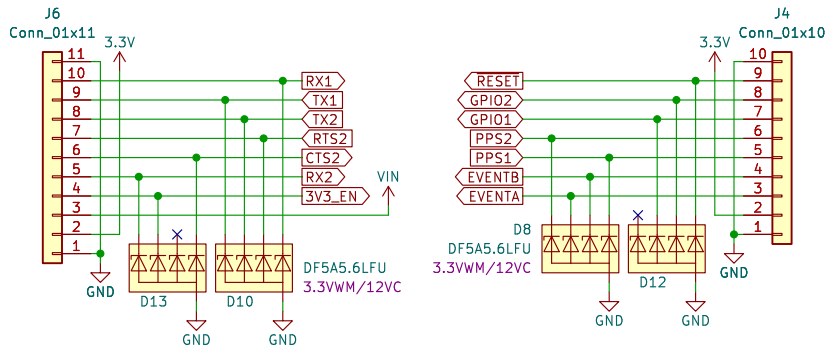
Power



mosaic-G5 P3 GNSS Receiver

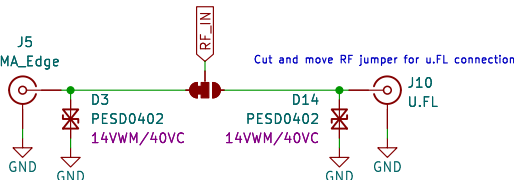


External PTH Connectors

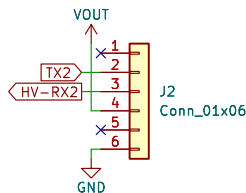


Microstrip (Coplanar Single Ended) 50Ω Calculation:
<https://jlcpcb.com/pcb-impedance-calculator>
Board thickness: 1.6mm, Layers: 4, Er: 4.4 (7628 Prepreg)
Dielectric Thickness (Layer 1 to 2): 8.28mil/0.2104mm (JLC7628)
Copper Thickness (1oz outer / 0.5oz inner): 1.38mil/0.035mm
Polygon Isolation: 7mil/0.1778mm
Trace Width: 11.1mil/0.2819mm

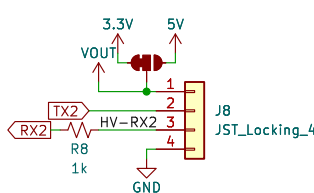
Antenna Connector



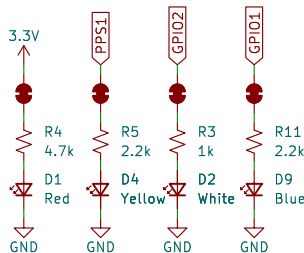
BlueSMiRF



Locking JST



LEDs



Designed by: P.C.
SparkFun Electronics
Sheet: /
File: SparkFun_GNSS_mosaic-G5_P3.kicad_sch

Title: mosaic-G5 P3 Allband GNSS Receiver Breakout

Size: USLedger | Date: 2025-04-23
KiCad E.D.A. 9.0.1

Rev: v01
Id: 1/1

